

KPC Secure Operational Data Pipeline

Strategic Business Case

Stanley Metone | Project Lead / Data Scientist | Team Apex Innovators

BOARD ASK: GO
Stage-gated implementation

1. Problem Statement

- Operational data from depot, pipeline and maintenance teams can arrive incomplete, duplicated or outside expected ranges, creating manual rework before it can be trusted.
- Staff, vehicle and location identifiers must be protected before the data is used for analysis, while weak release controls can create avoidable operational and compliance risk.
- Business impact: analyst time is consumed by clean-up, decisions are delayed, and errors can reach reporting or alerting workflows.

2. Proposed Solution

- Create one controlled data route that automatically checks incoming records, removes or blocks bad records, and masks defined sensitive identifiers.
- Flag abnormal pressure patterns early and present decision-ready status to managers in one dashboard.
- Only release tested versions after automated security and health checks, with a safe rollback if a release fails.

3. Financial Impact - planning case to validate with KPC data

KES 5.9M

Net annual savings

171%

Year-1 ROI

3.7 months

Payback period

Savings math

25,000 records/week x 5% needing manual review x 4 minutes/record / 60 x KES 1,500/hour x 52 weeks = KES 6.50M gross annual labour cost avoided. Less KES 0.60M annual support = KES 5.90M net recurring savings.

Investment and return math

One-time implementation: KES 1.80M. Year-1 total cost = KES 1.80M + KES 0.60M = KES 2.40M. ROI = (KES 6.50M - KES 2.40M) / KES 2.40M = 171%. Payback = KES 1.80M / (KES 5.90M / 12) = 3.7 months.

4. Top Risks & Mitigation

Risk 1 - Financial assumptions do not match real operations

Use an 8-week pilot on representative KPC data. Recalculate savings using actual record volumes, review time and labour cost before full rollout.

Risk 2 - Automation misses a privacy, quality or alerting issue

Keep a human override, verify 100% masking of defined identifiers, track false alerts, restrict access, and retain automatic rollback for failed releases.

5. Recommendation: GO - approve a stage-gated implementation

Approve an implementation ceiling of KES 1.80M, released in stages, with an 8-week pilot as the first decision gate. Proceed to scale only if the pilot confirms at least 95% accepted records, 100% masking of defined sensitive fields, less than 10% noisy alerts, and reliable release health checks.

Decision required: executive sponsor + access to representative operational data + staged budget approval.